

IX: Gaussian Splatting

3D CV

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In the Previous Episode

- Neural Radiance Fields (NeRF) for novel view synthesis
 - Volumetric representations for 3D data
 - Volumetric rendering
- Improvements & applications of NeRFs



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Neural Radiance Fields

Representing a Scene with Neural Fields

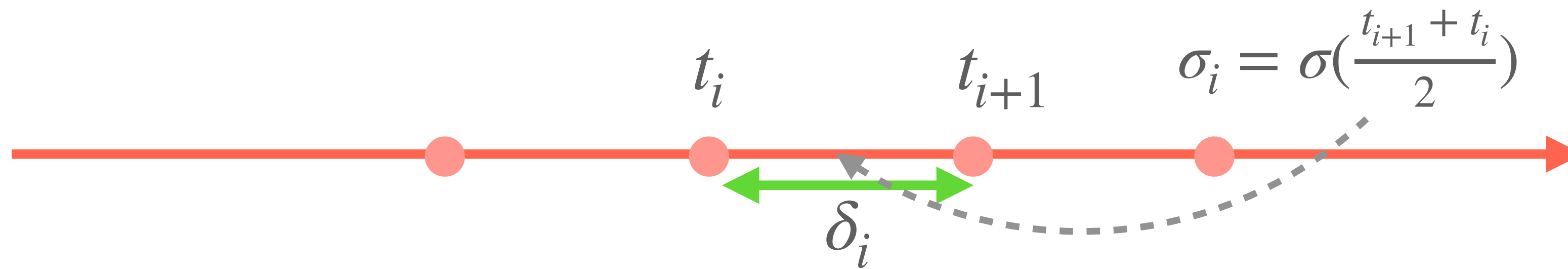
- Represent a scene with two fields
 - Density: $\sigma(x) : \mathbb{R}^3 \rightarrow \mathbb{R}^+$
 - Radiance: $C(x, d) : \mathbb{R}^3 \times S^2 \rightarrow \mathbb{R}^3$
- Density represents is related to opacity
- Radiance represents color of a point
- Architecture: MLP + positional embeddings



Computing Pixel Color

- Density $\sigma(t) \in [0, +\infty)$ is related to opacity at t
- Divide the ray with points $t_1, \dots, t_n$ and define

$$\alpha_i = 1 - \exp(-\sigma_i \delta_i)$$



- Expected color along a ray is given by

$$C = \sum_i c(t_i) \cdot \alpha_i \prod_{j < i} (1 - \alpha_j)$$

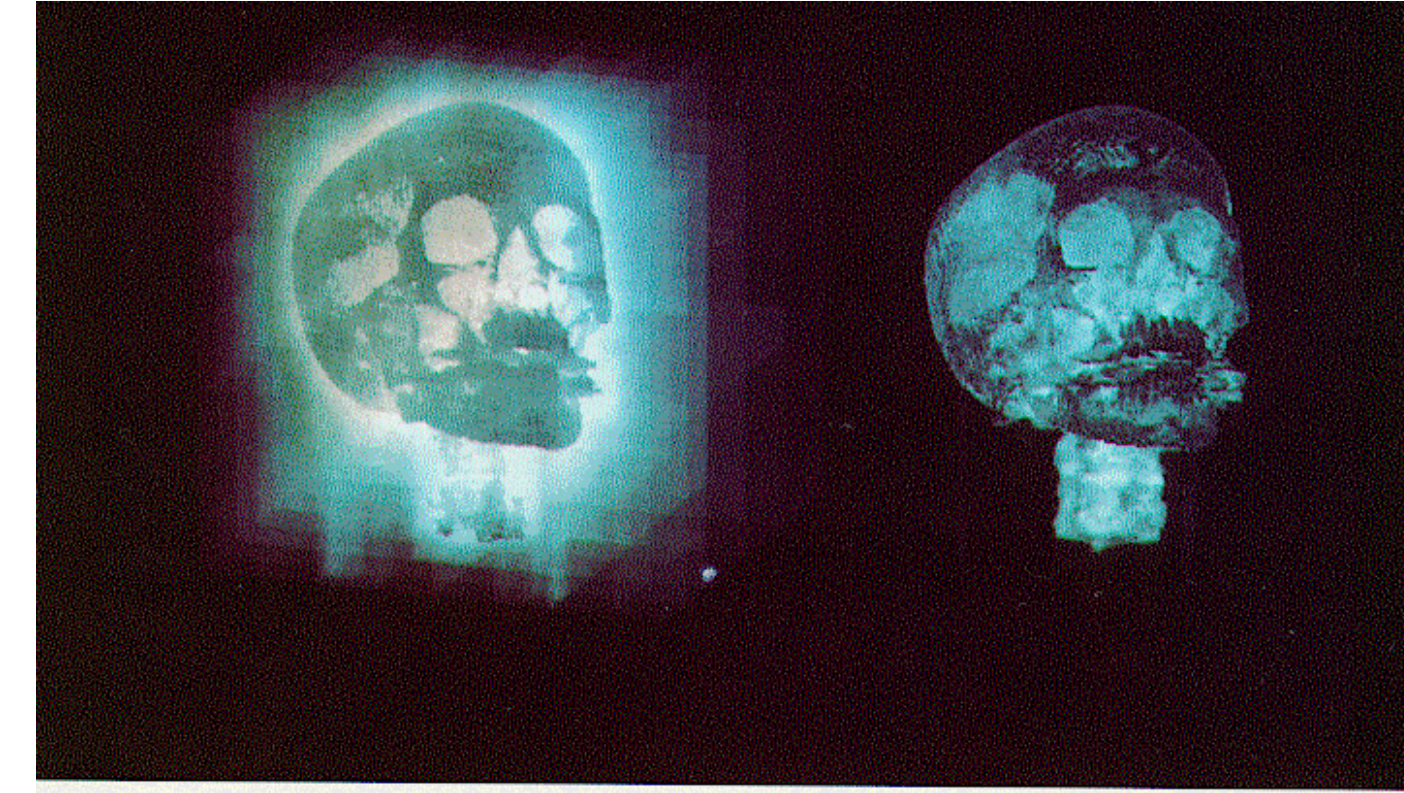
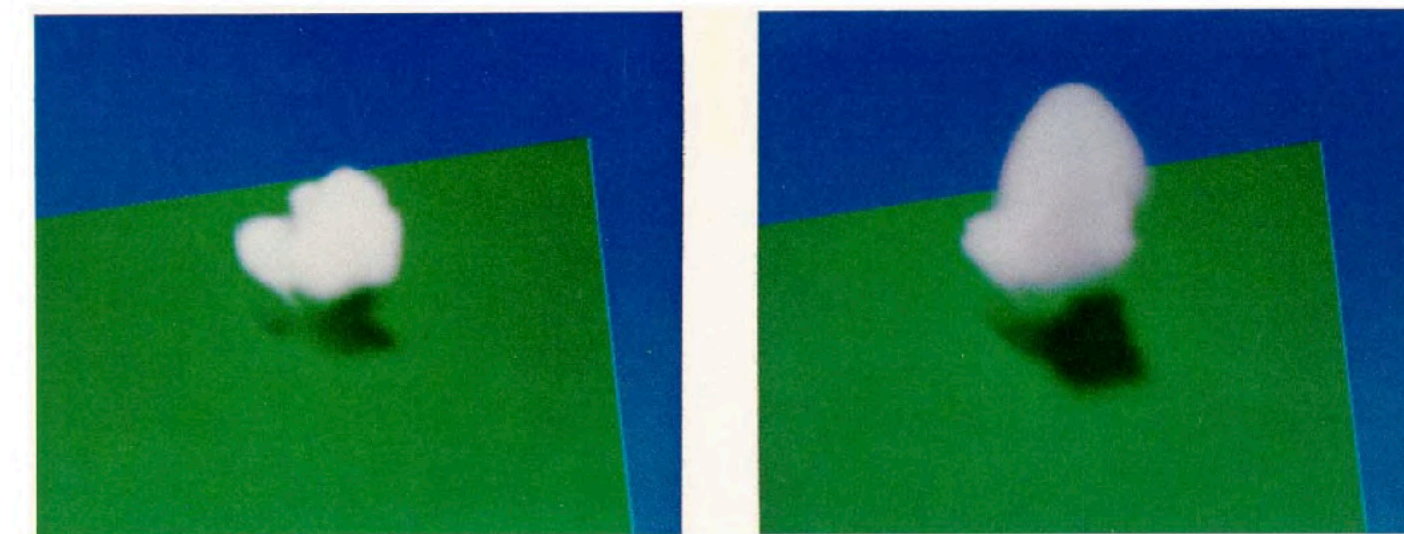
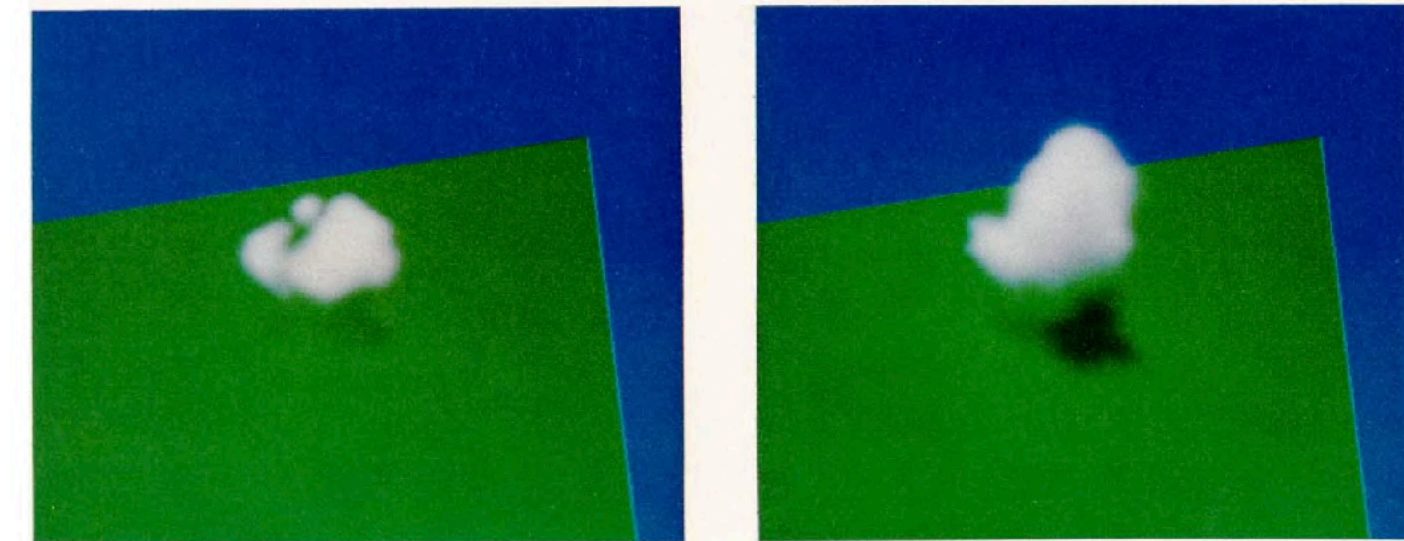


Fig. 11. Costs of rendering Figure 8 using hierarchical enumeration and adaptive termination.



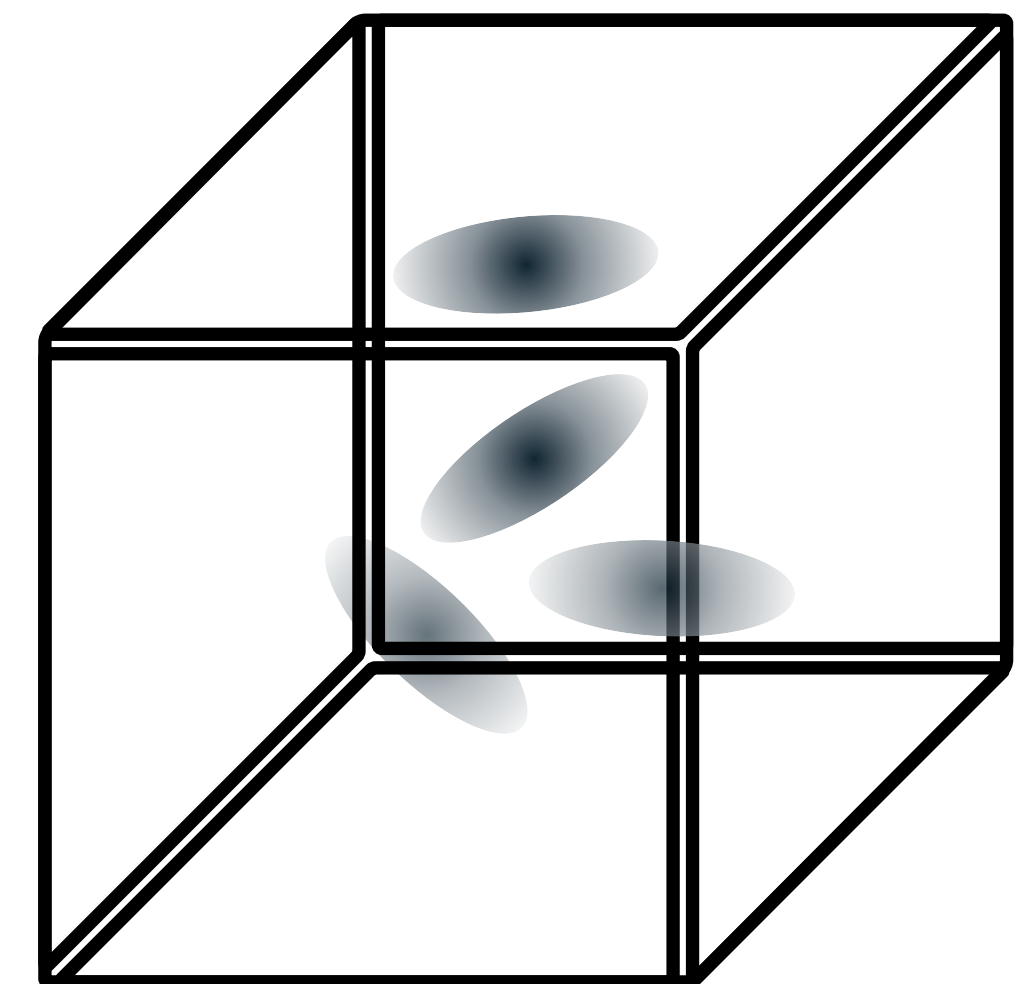
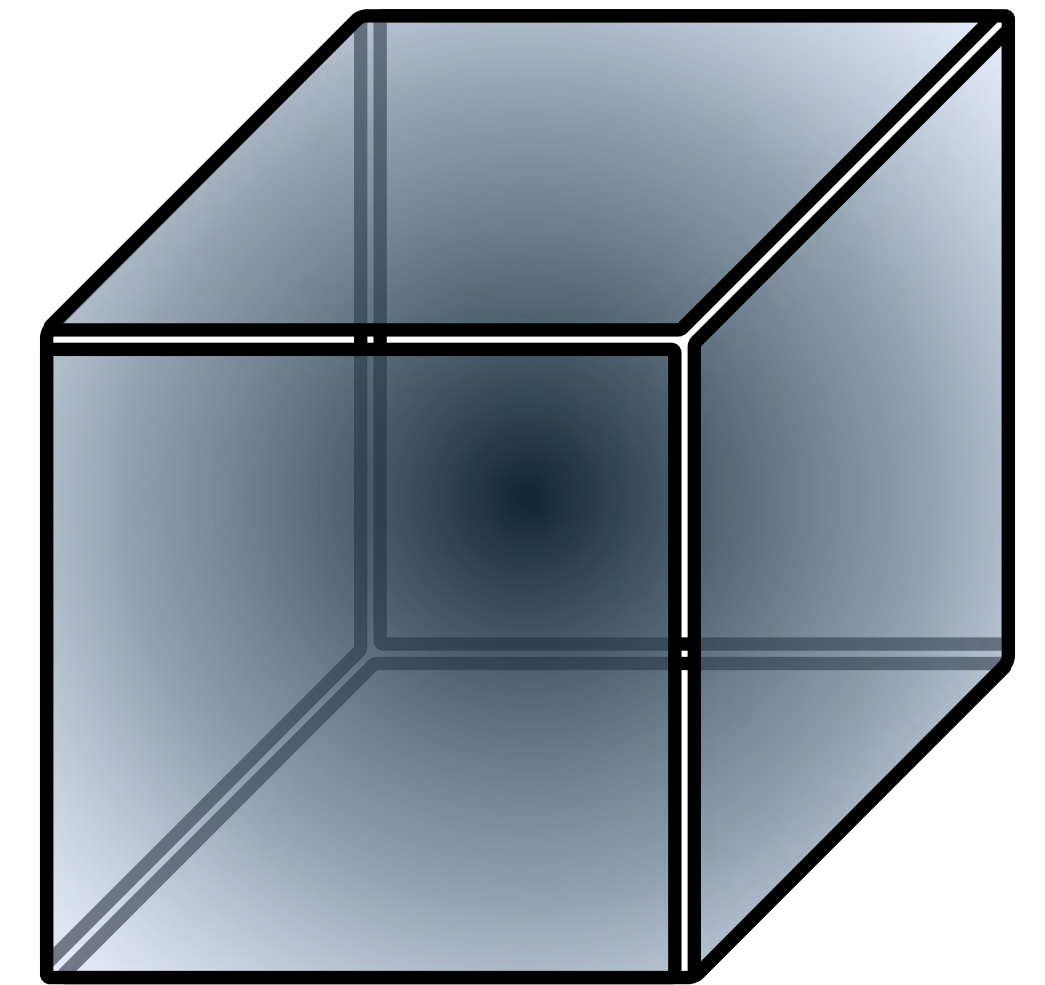
Computational Overhead

- Notorious rendering time even with high-end GPUs
 - MipNeRF: 0.07 FPS
 - Instan-NGP: 9.0 FPS
- Today: gaussian splatting
 - Alternative solution allowing real-time rendering
 - Gaussian Splatting: 100+ FPS

Gaussian Splatting

Idea: Volumetric Rendering + Rasterisation

- Neural Radiance Fields employ a **dense** field parameterisation
- Rendering algorithm runs independently for each pixel
- Gaussian splatting employs a **sparse** parameterisation
 - Similar to meshes, scene consists of primitive shapes
 - Unlike meshes, these primitives have volume
 - Volumetric rendering algorithm based on **rasterization**







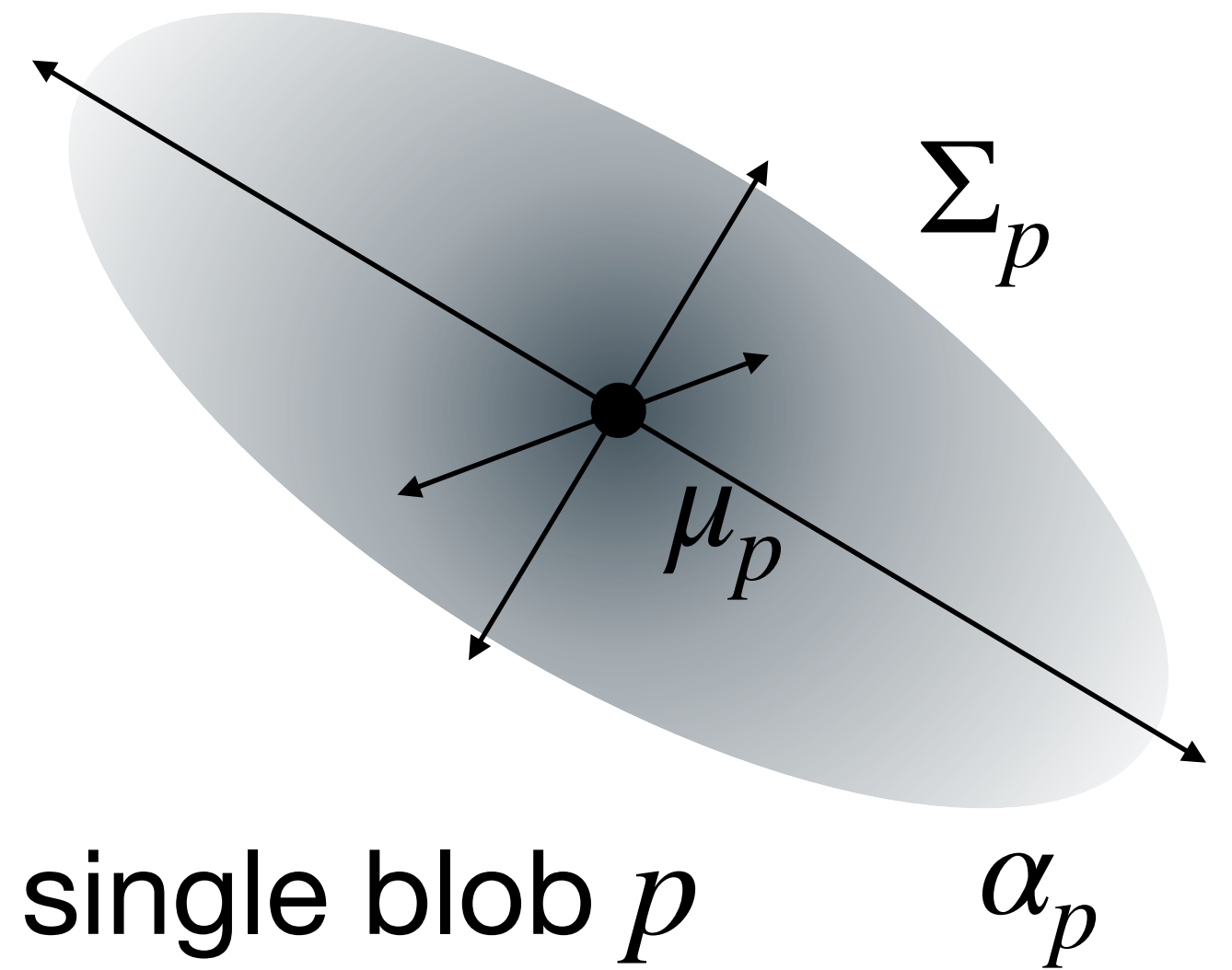


Representing Scene with Gaussians

- We will fill the volume with multiple tiny blobs
- In NeRFs we relied on a density field $\sigma(x)$
- Here we will define opacity $\alpha(x) = 1 - \exp(\sigma(x)\Delta x)$ of a single blob p

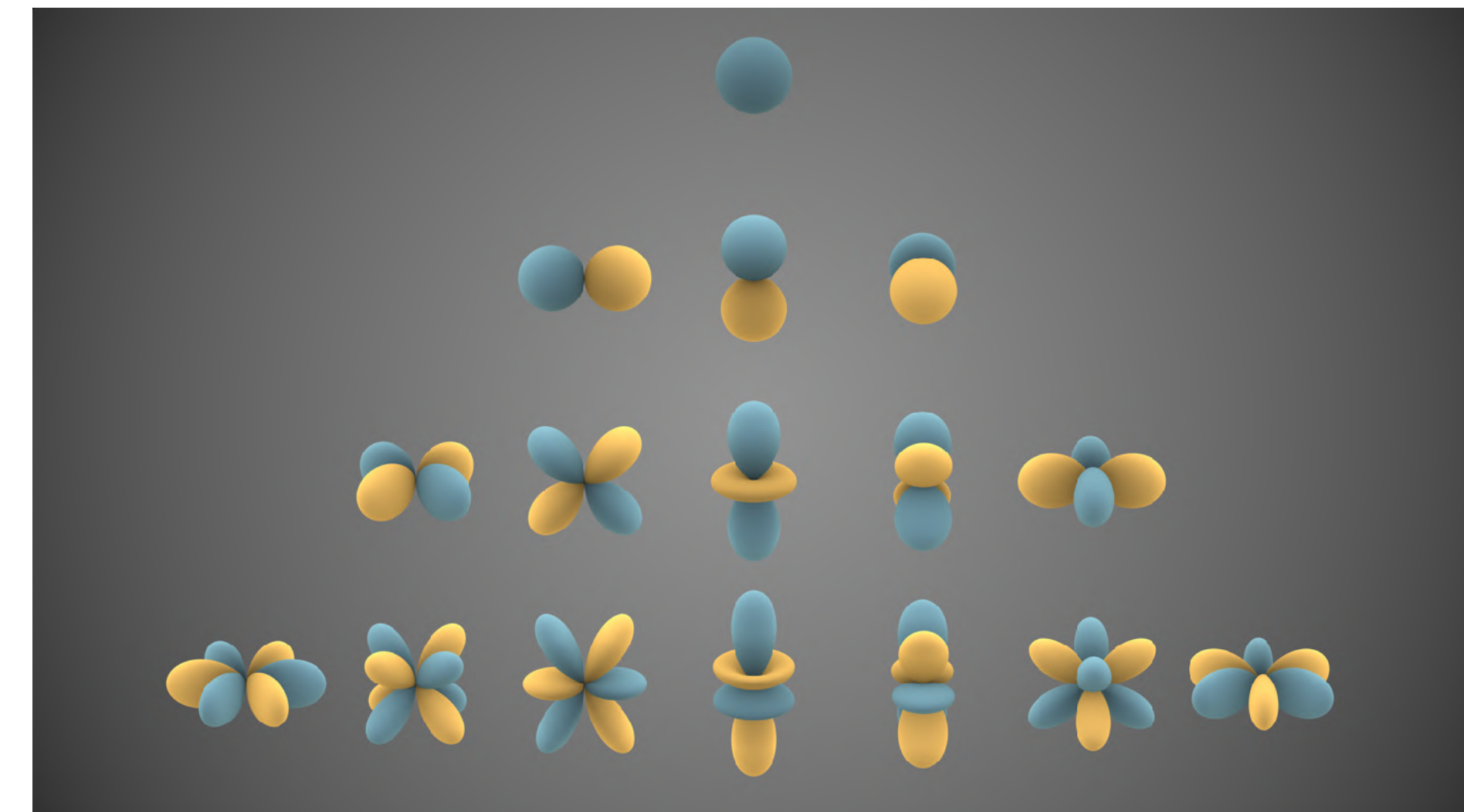
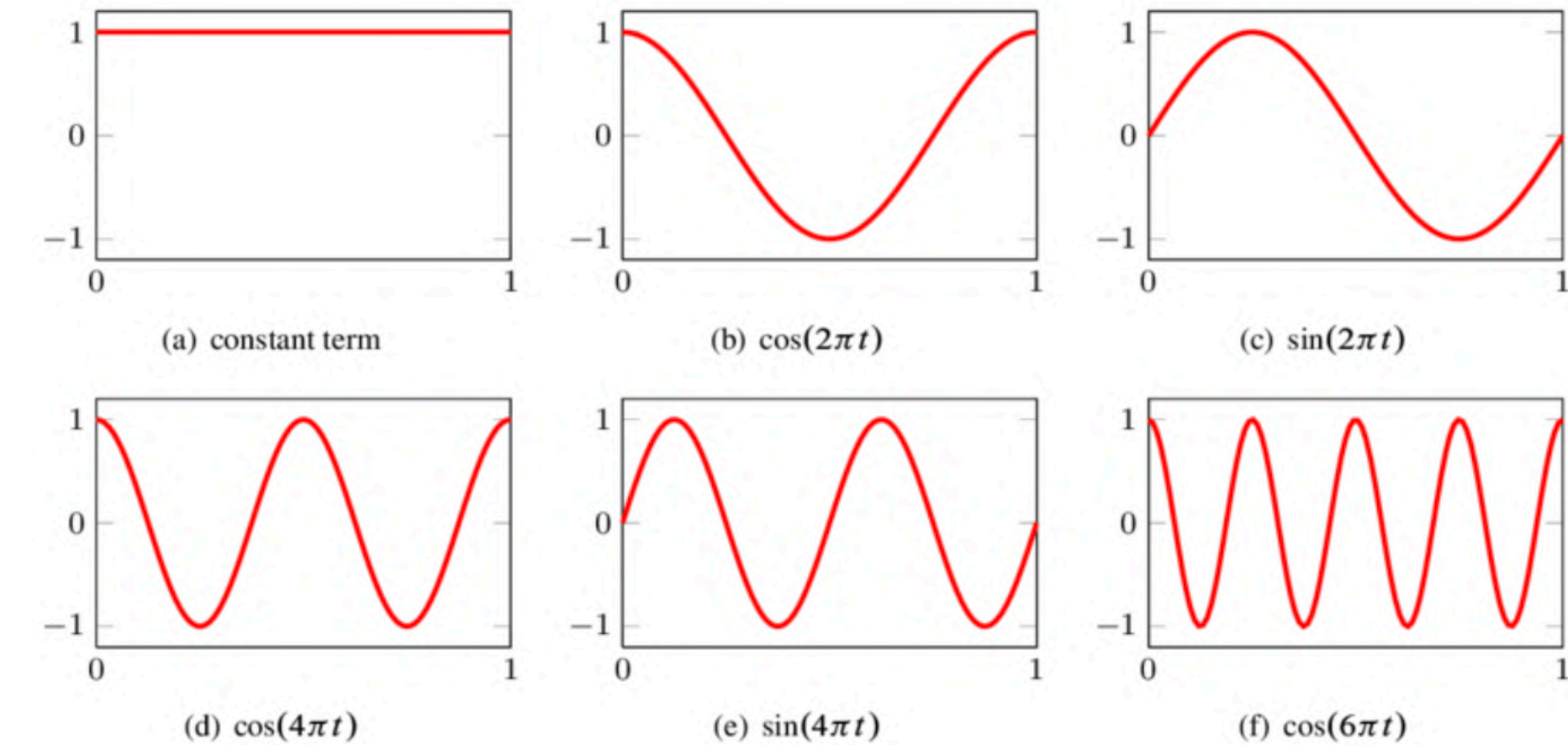
$$\alpha(x) = \alpha_p \exp \left(-\frac{1}{2}(x - \mu_p)^T \Sigma_p^{-1} (x - \mu_p) \right)$$

- Parameters $\alpha_p, \mu_p, \Sigma_p$ define opacity, position, and shape of the blob respectively



Parameterizing Radiance with SH

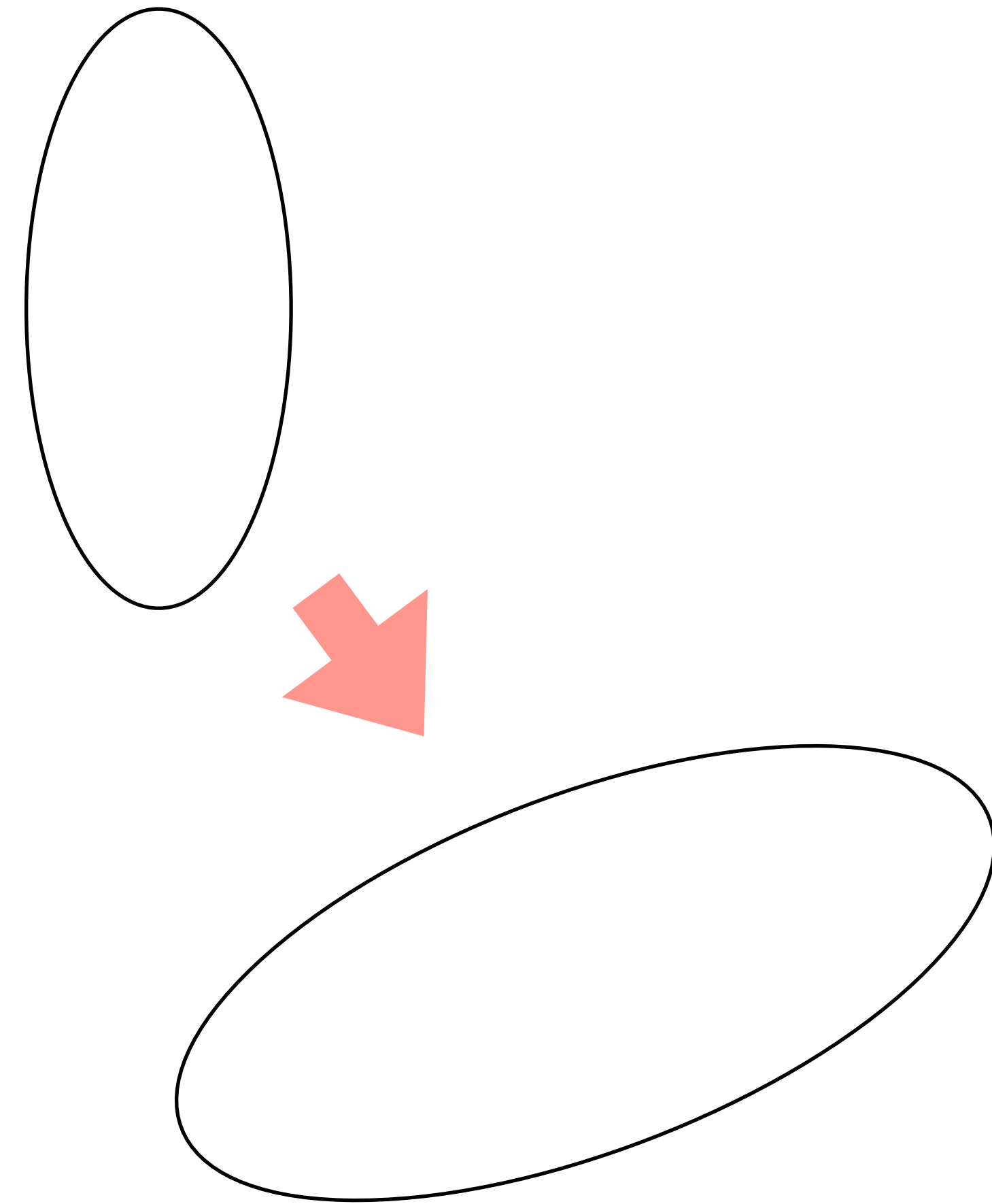
- Each blob radiates light
- Radiance function depends on ray direction
- Omit dependence on position within the blob
- Similar to $\{\sin(k\pi x), \cos(k\pi x)\}_{k \in \mathbb{N}}$ on $[0,1]$, **spherical harmonics** define an orthogonal functional basis for S^2 functions
 - *Physicists: harmonics are solutions to wave equation*
 - *Mathematicians: harmonics are eigenfunctions of Laplace operator*



Triangles vs Gaussians

Affine Transformations

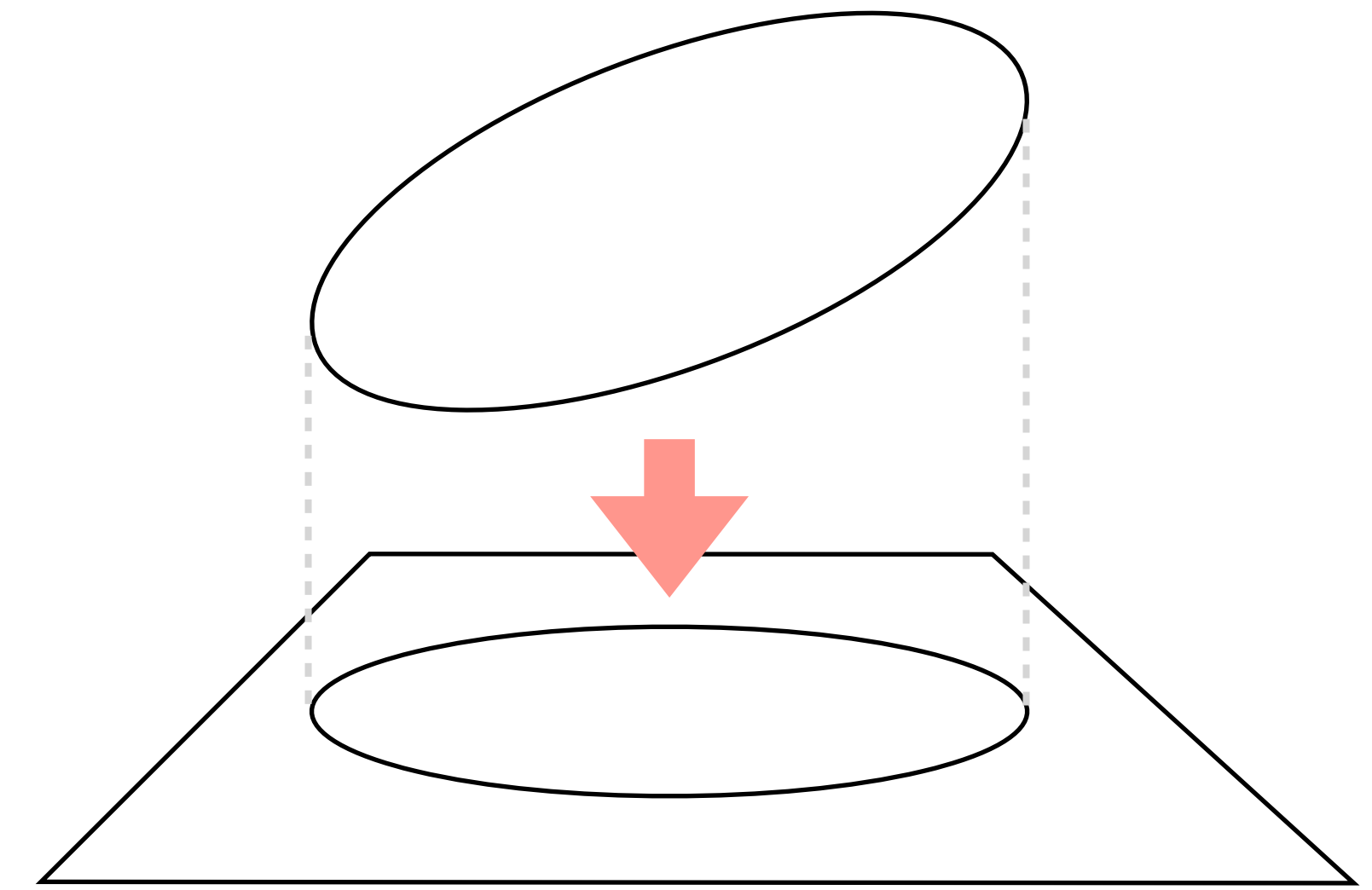
- Triangles have multiple convenient properties
- A projective transformation of a triangle is still a triangle
- An affine transformation of a Gaussian is still a Gaussian
 - Let $\xi \sim \mathcal{N}(\mu, \Sigma)$ and $\phi = A\xi + b$
 - Then $\phi \sim \mathcal{N}(A\mu + b, A\Sigma A^T)$
- We will use this property for *world-to-camera* transformation



From 3D to 2D

Marginalization

- Marginal distribution of a Gaussian is a Gaussian



- For a r.v. $\begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix} \sim \mathcal{N}(\mu, \Sigma)$ distribution of $\begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix}$ is $\mathcal{N} \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right)$
- From graphics standpoint, vector $\begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix}$ is an orthographic projection of $\begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix}$

Perspective Projection with Gaussians

Projection

- Perspective projections are non-linear in Euclidian space

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \rightarrow \begin{pmatrix} f \frac{x}{z} \\ f \frac{y}{z} \end{pmatrix}$$

- We can use the Jacobian at $x = \mu_p$ of projection as a substitute

$$J = \begin{pmatrix} \frac{f}{z} & 0 & -\frac{fx}{z^2} \\ 0 & \frac{f}{z} & -\frac{fy}{z^2} \end{pmatrix}$$

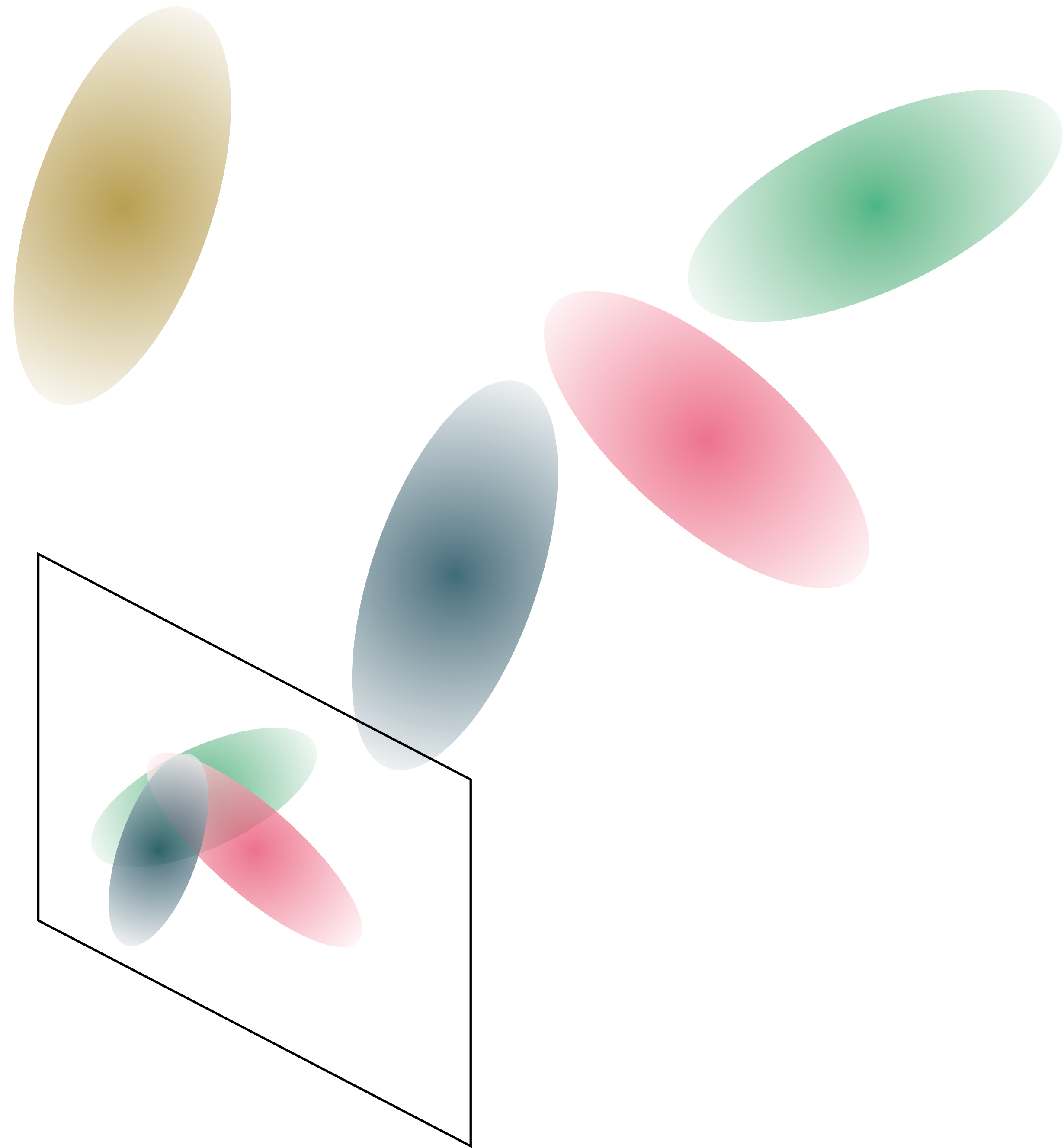
- In practice, we end up with 2D Gaussian with $\Sigma' = JW\Sigma W^T J^T$, where W is world-to-cam matrix

Rendering Algorithm

Rendering multiple splats

- Culling
 - Exclude splats outside of the frame
- Render each splat from closest to farthest
 - Blend frame with alpha-compositing

$$C = \sum_i C_i \alpha_i \prod_{j < i} (1 - \alpha_j)$$



Pseudo-Code

Why Rasterization is so Efficient?

```
for each pixel # NeRF, 10e6 pixels
```

```
    for each segment on a ray # 10e2 ray segments
```

```
        compute & accumulate radiance
```

```
for each splat # Gaussian splatting, 10e4 splats
```

```
    for each pixel on a splat # 10e2 pixels in a splat
```

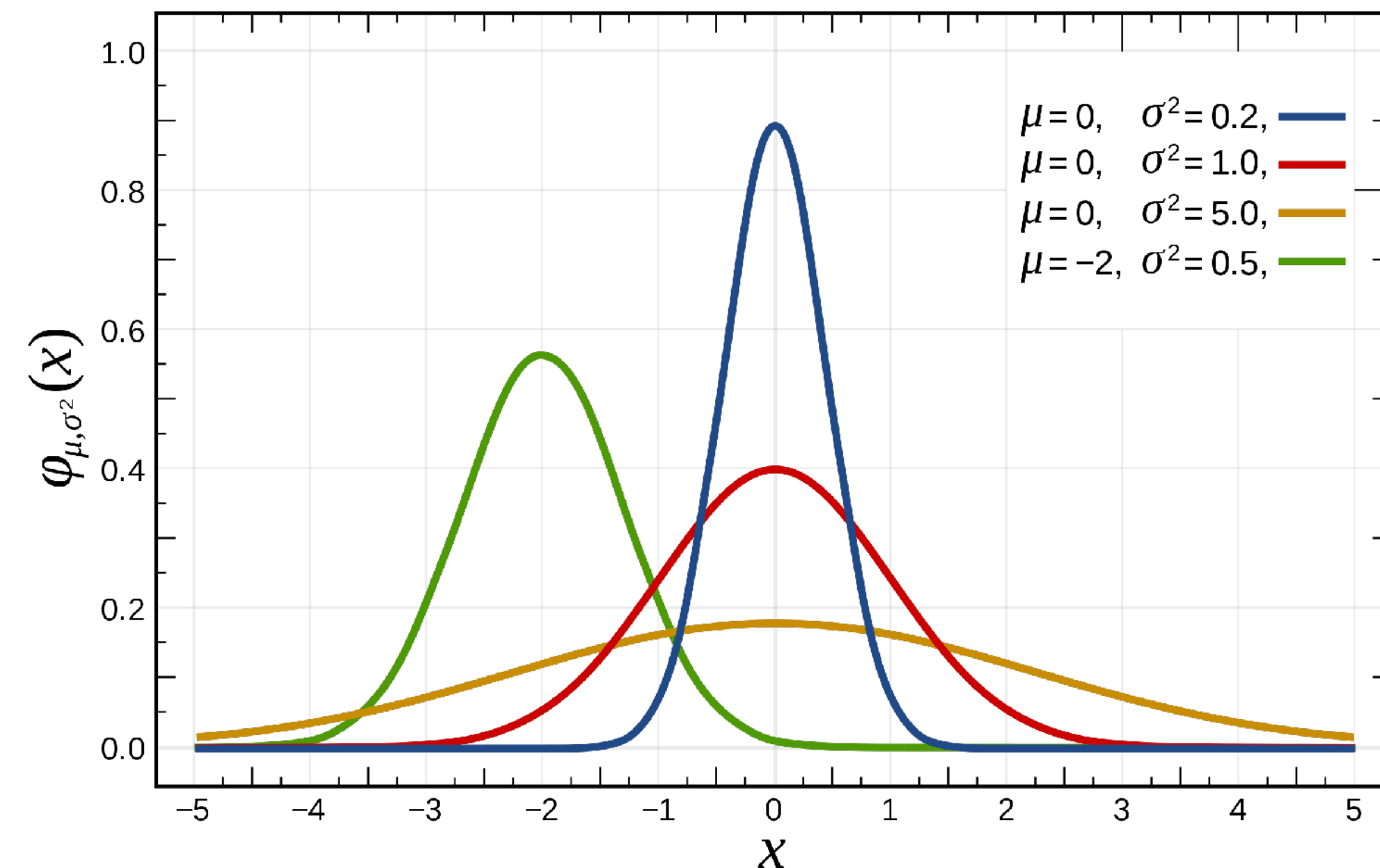
```
        compute & accumulate radiance
```


Fitting Gaussian Splats

A Bag of Tricks

- Unlike triangles in meshes, Gaussian splats have *soft* edges
- As a result, splats amenable to optimization
- However, gradient vanishes far from the mean
- Photometric loss has multiple local minima

$$\|I_{gt} - I_{rendered}\|^2$$



Fitting Gaussian Splats

Initialisation with SfM

PSNR 20.90

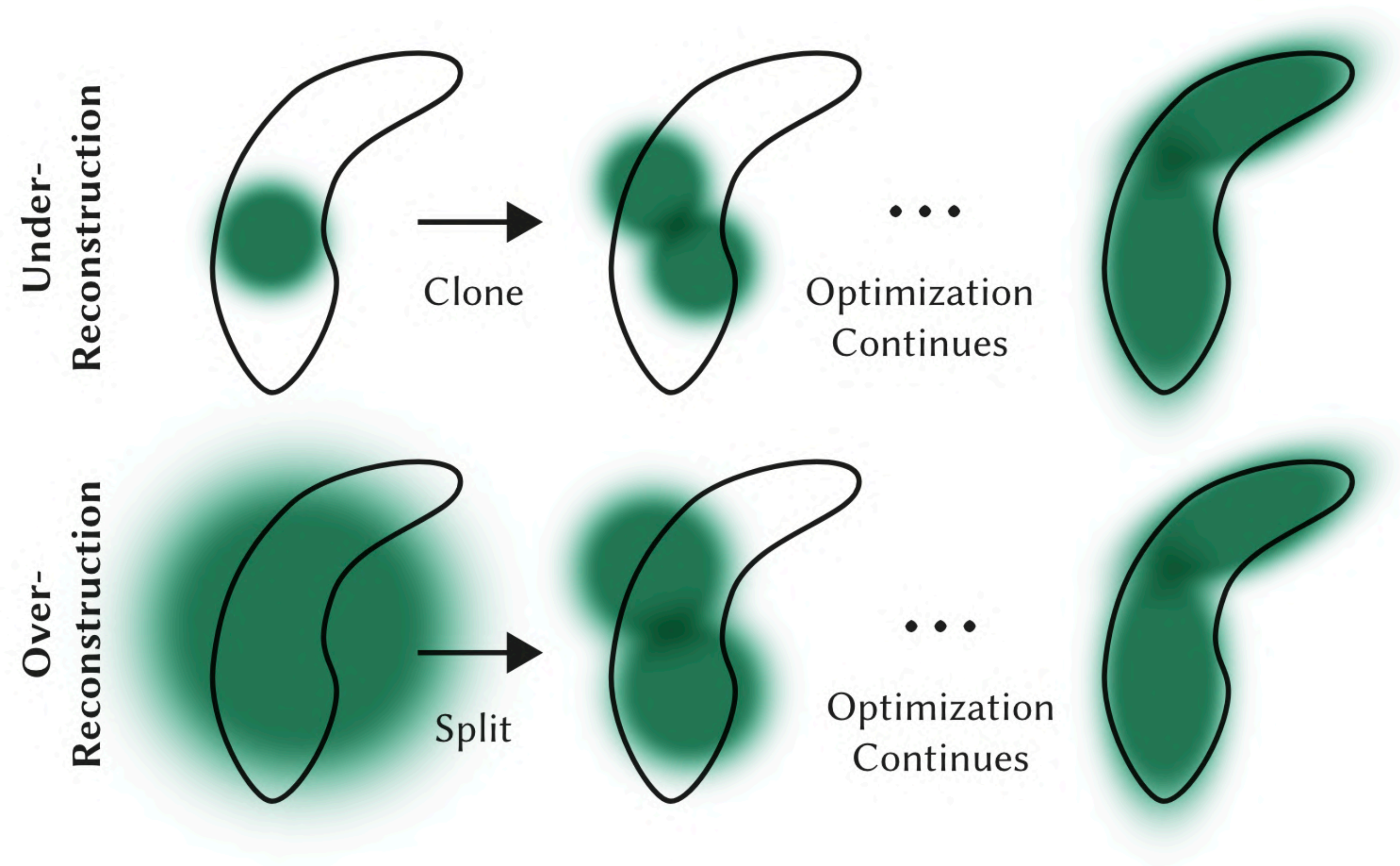


PSNR 25.82



Fitting Gaussian Splats

Adaptive Control of Gaussians



Overview

- Pros: efficiency
 - Blazingly fast inference
 - Training is also relatively fast
- Cons: overparameterized representation
 - Local optima, PSNR does not match NeRFs
 - Naive representation takes up to 1GB of memory

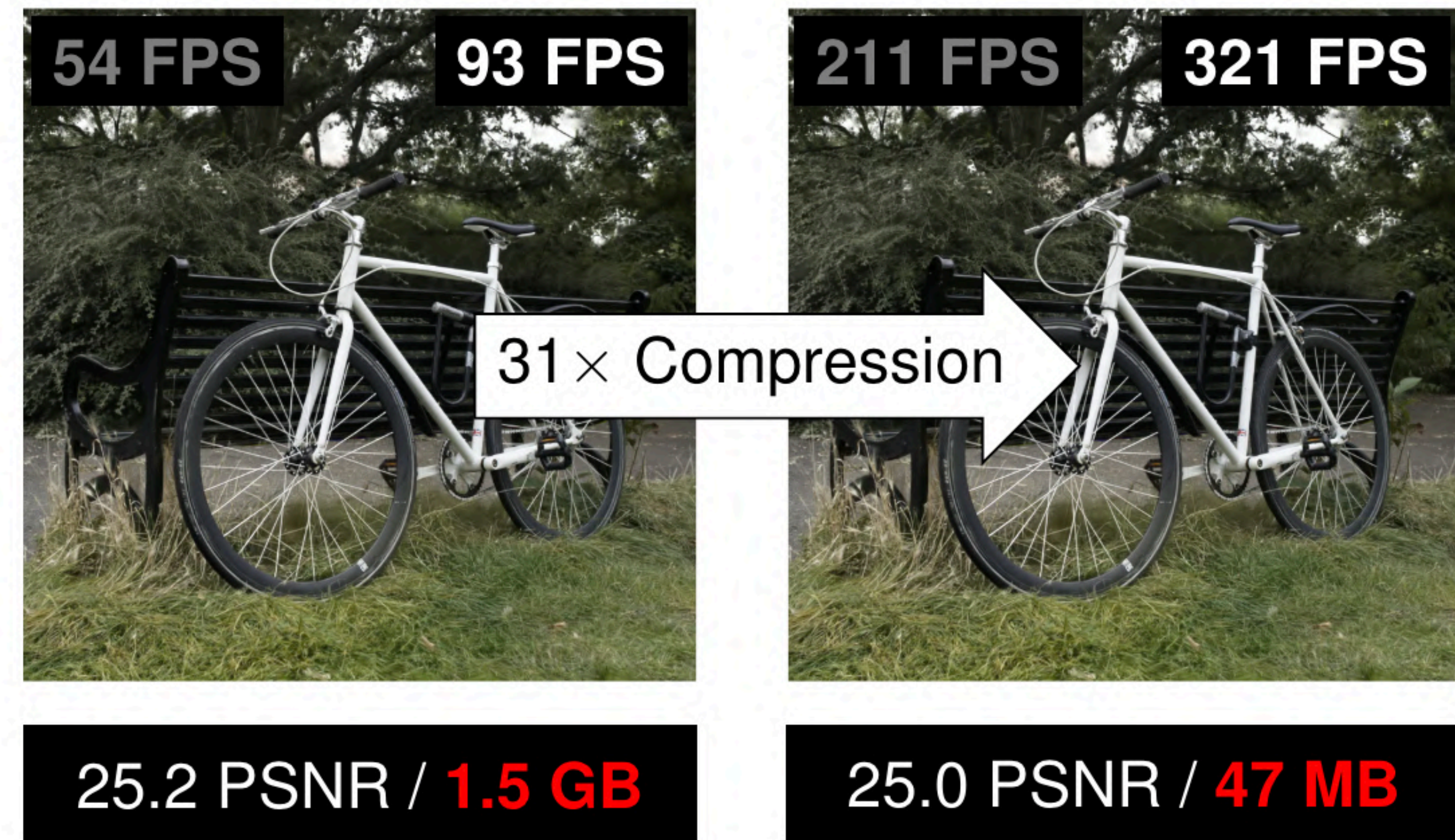
Compression

Memory Footprint

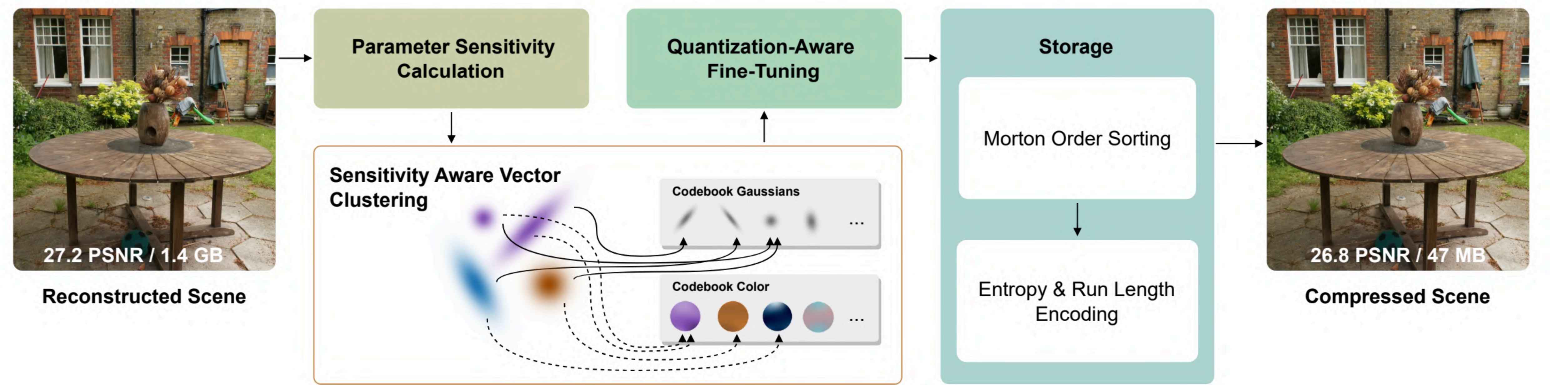
- High-quality scene contains 30k splats
 - Shape is parameterised with mean (3 floats) and covariance (6 floats)
 - Radiance is parameterized with α (1 float) and $16 \times 3 = 48$ SH coefficients
- Approximately $7 \cdot 10^6$ bytes
 - **Color** constitutes **82%** of weights
 - Gaussian **covariance** constitutes **10%** of weights
- Differs from NeRFs by two orders of magnitude

Room for Optimization

- Memory footprint can be reduced with standard solutions
 - Quantization
 - Low-precision parameterisation
 - Compression

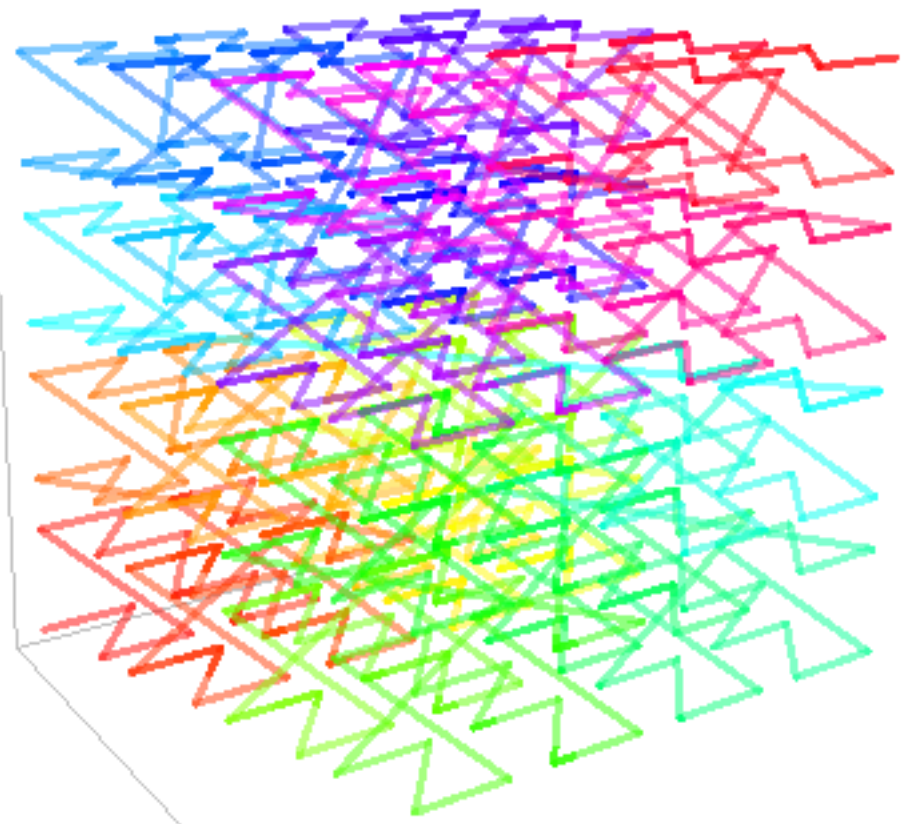


Compression Pipeline



		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	SIZE $\downarrow$
Color	1024	26.95(-0.67)	0.80(-0.02)	0.24(+0.03)	28.47
	2048	26.95(-0.62)	0.80(-0.02)	0.24(+0.03)	28.65
	4096	26.98(-0.63)	0.80(-0.02)	0.24(+0.03)	28.80
	8192	27.00(-0.58)	0.80(-0.02)	0.24(-0.03)	28.92
Gaussian	1024	26.95(-0.79)	0.80(-0.02)	0.24(+0.03)	28.14
	2048	26.97(-0.80)	0.80(-0.02)	0.24(+0.03)	28.45
	4096	26.98(-0.63)	0.80(-0.02)	0.24(+0.03)	28.80
	8192	26.97(-0.60)	0.80(-0.02)	0.24(+0.03)	29.06

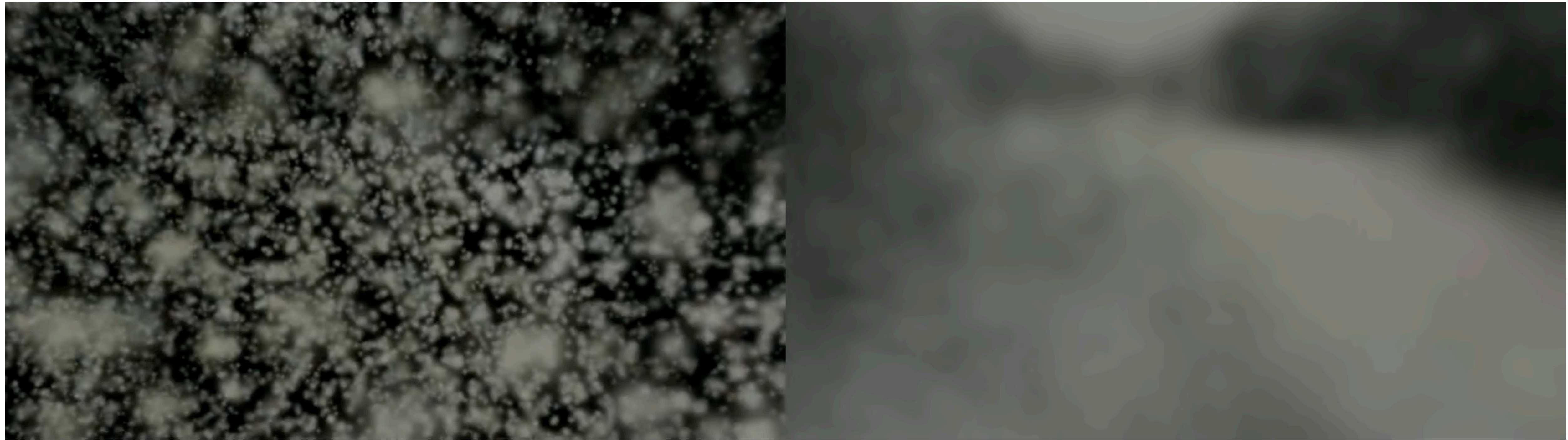
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	SIZE $\downarrow$
baseline	27.179	0.861	0.115	1379.99
+ Pruning	27.083	0.856	0.118	1217.25
+ Color Clustering	25.941	0.818	0.178	278.41
+ Gaussian Clustering	25.781	0.811	0.186	164.15
+ QA Finetune	26.746	0.844	0.144	86.69
+ Encode	26.746	0.844	0.144	58.40
+ Morton Order	26.746	0.844	0.144	46.57



Technology is Still In Development

Do We Really Need a Proper Initialisation?

3D Gaussian Splatting as Markov Chain Monte Carlo



Digital Image Processing Details Matter

Exposure Variability Example



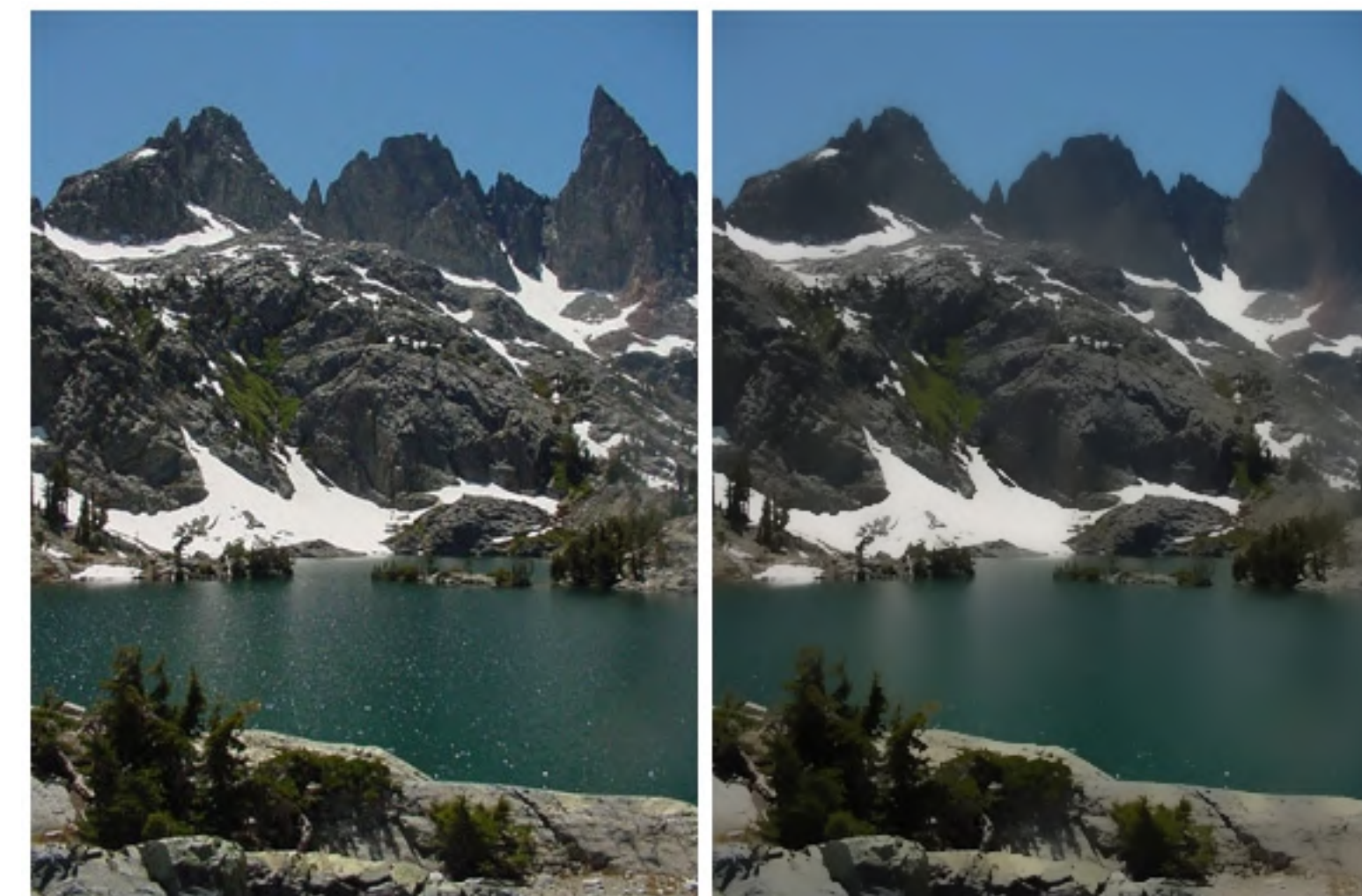
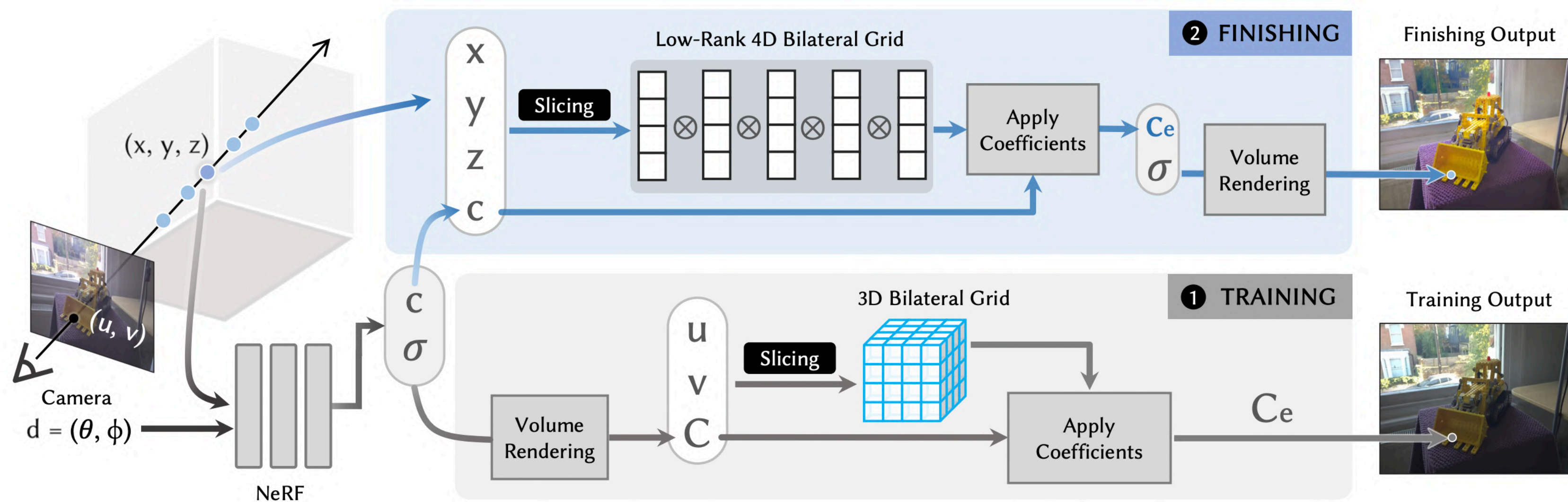
Digital Image Processing Details Matter

Pixel Values $\neq$ Radiance

- Raw sensor data undergoes multiple transformations:
 - Tone-mapping & white balancing
 - Denoising
 - Gamma-correction

Data Adjustments with Bilateral Grids

- Key idea: learn image processing adjustments for each frame during training
- Image adjustments are local: 2D grid of linear transformations
- Models bilateral filters used in image processing to preserve edges



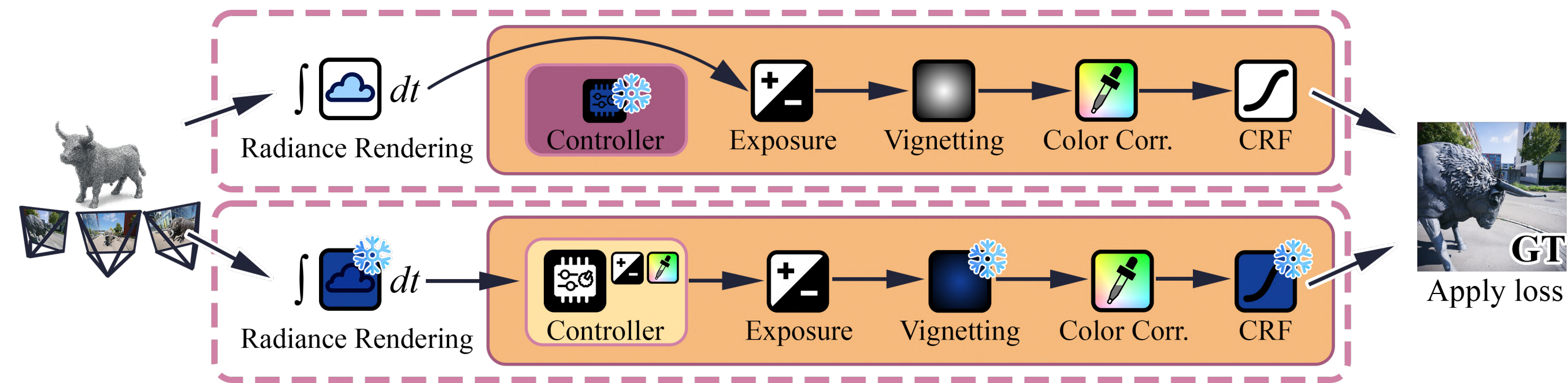
Bilateral filter

Data Adjustments with Bilateral Grids



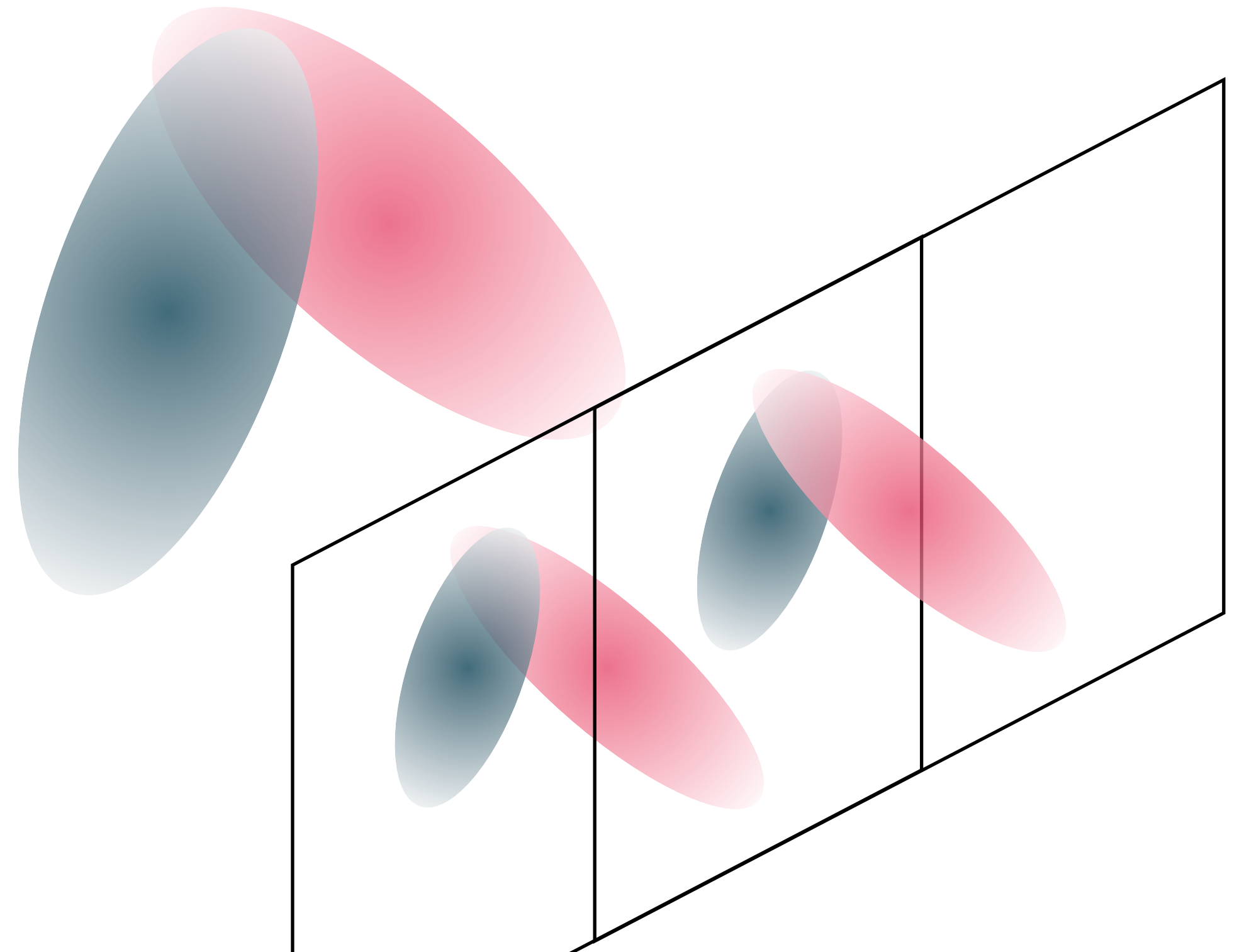
Physically-Plausible Processing Simulations

Distill Sensor Adjustments into a Separate Module



Rendering Algorithm Shortcomings

- Distance-based splat ordering results in popping artefacts
- Formula for projected splat shape is a linear approximation



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- Formula for projected splat shape is a linear approximation
- Pinhole model is limited
 - Data comes from fish-eye lenses
 - Often contains rolling shutter effects



Rendering Algorithm Shortcomings

- Distance-based splat ordering results in popping artefacts
- Formula for projected splat shape is a linear approximation
- Pinhole model is limited
 - Data comes from fish-eye lenses
 - Often contains rolling shutter effects
- Ray tracing solves these issues, but too slow even with hardware acceleration
- Further rasterisation improvements mitigates those issues (3DGUT)

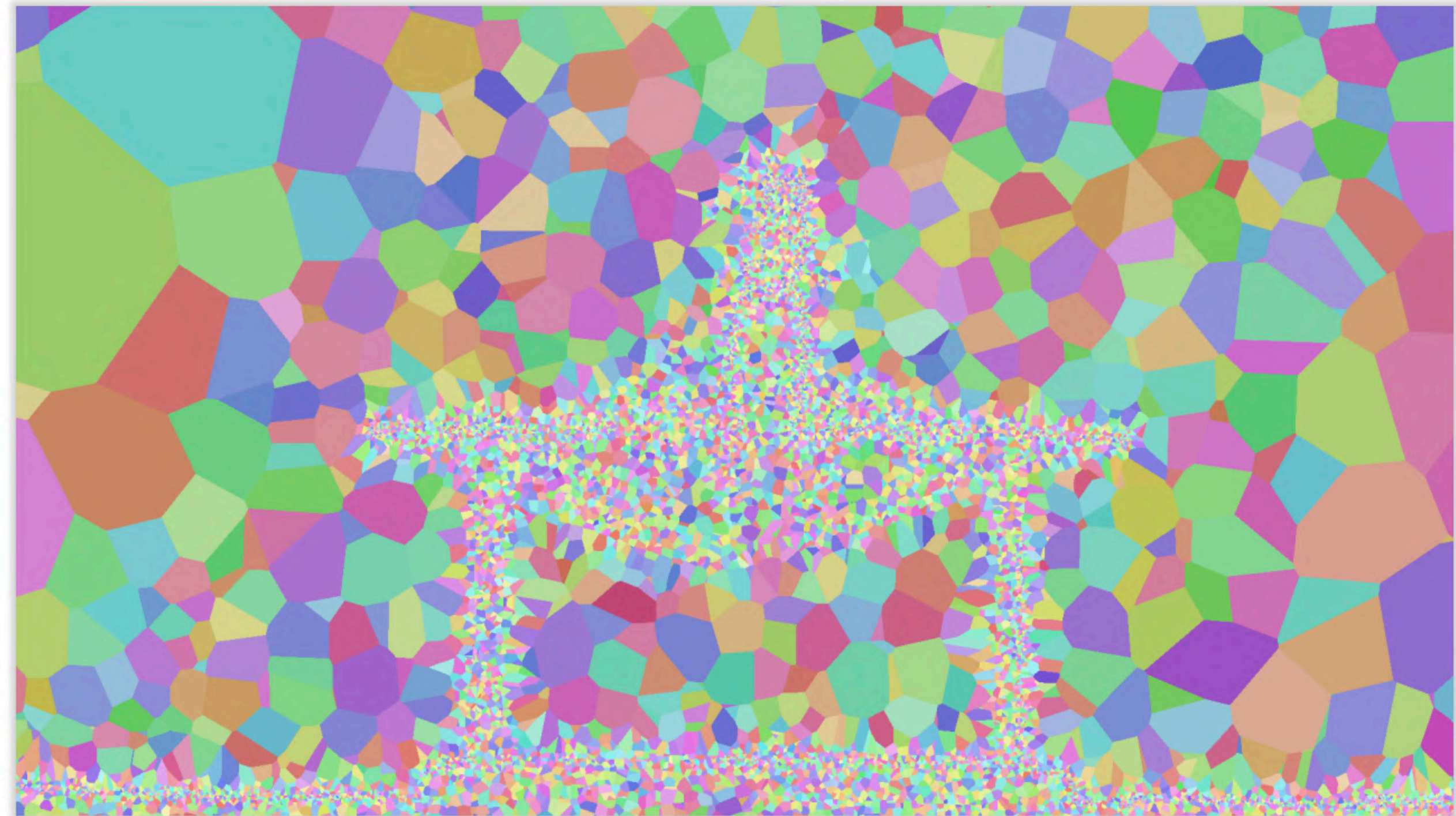


<https://research.nvidia.com/labs/toronto-ai/3DGRT/>

<https://research.nvidia.com/labs/toronto-ai/3DGUT/>

Beyond Gaussians

Voronoi Diagrams



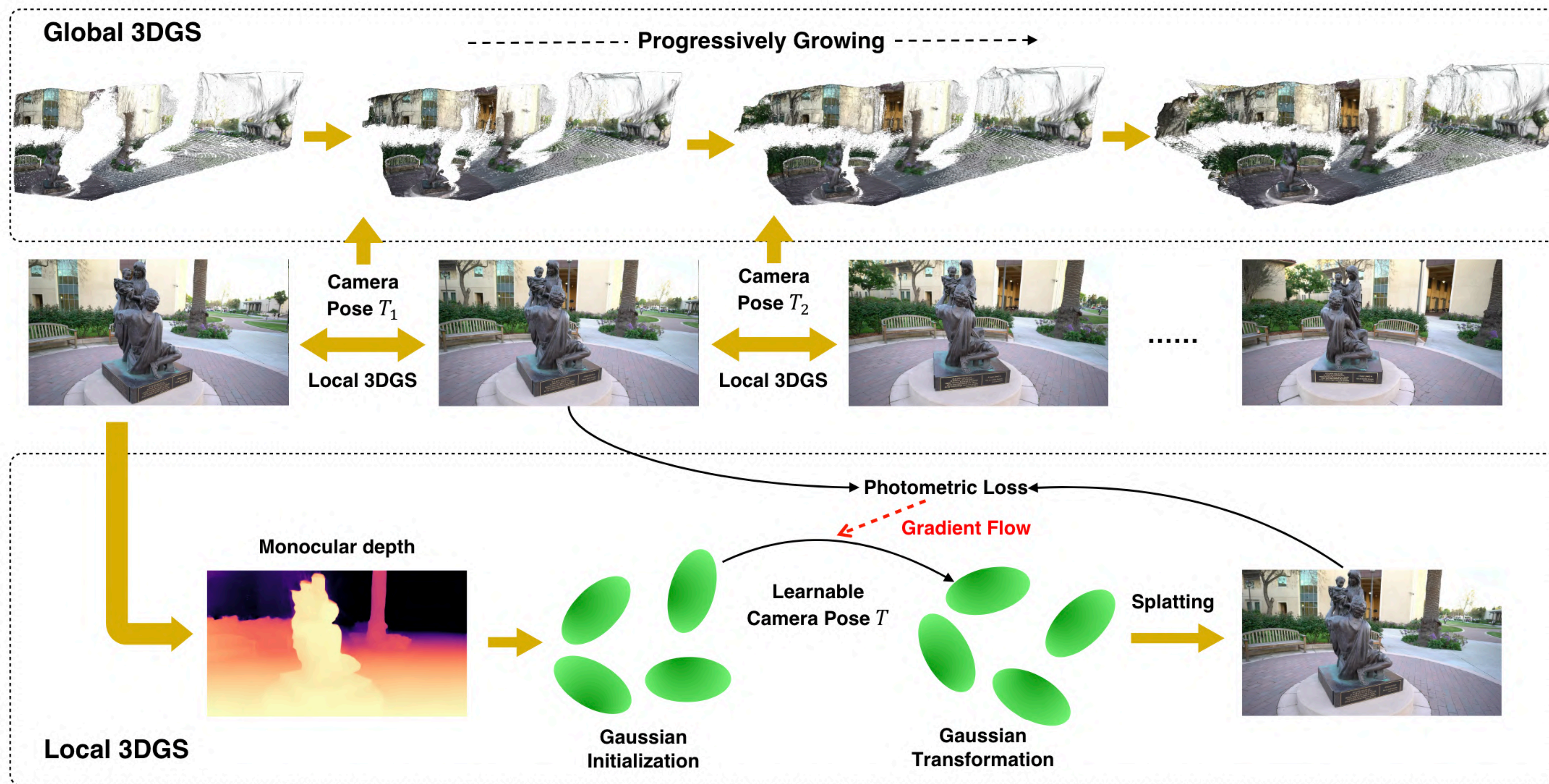
Gaussian Splats without Colmap

Do We Need Camera Positions?

- Novel view synthesis still relies on SfM (COLMAP)
- NeRF allow optimizing camera positions
 - BARF, Nope-NeRF, etc.
- Learning positions from scratch is still a challenge

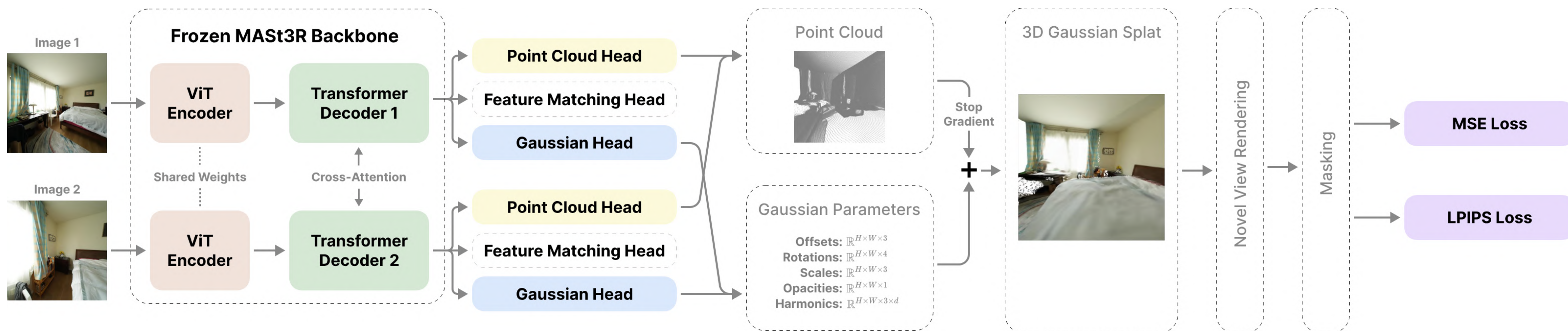


COLMAP-Free 3D Gaussian Splatting



Mast3R + Gaussian Splatting

InstantSplat, Splatt3R

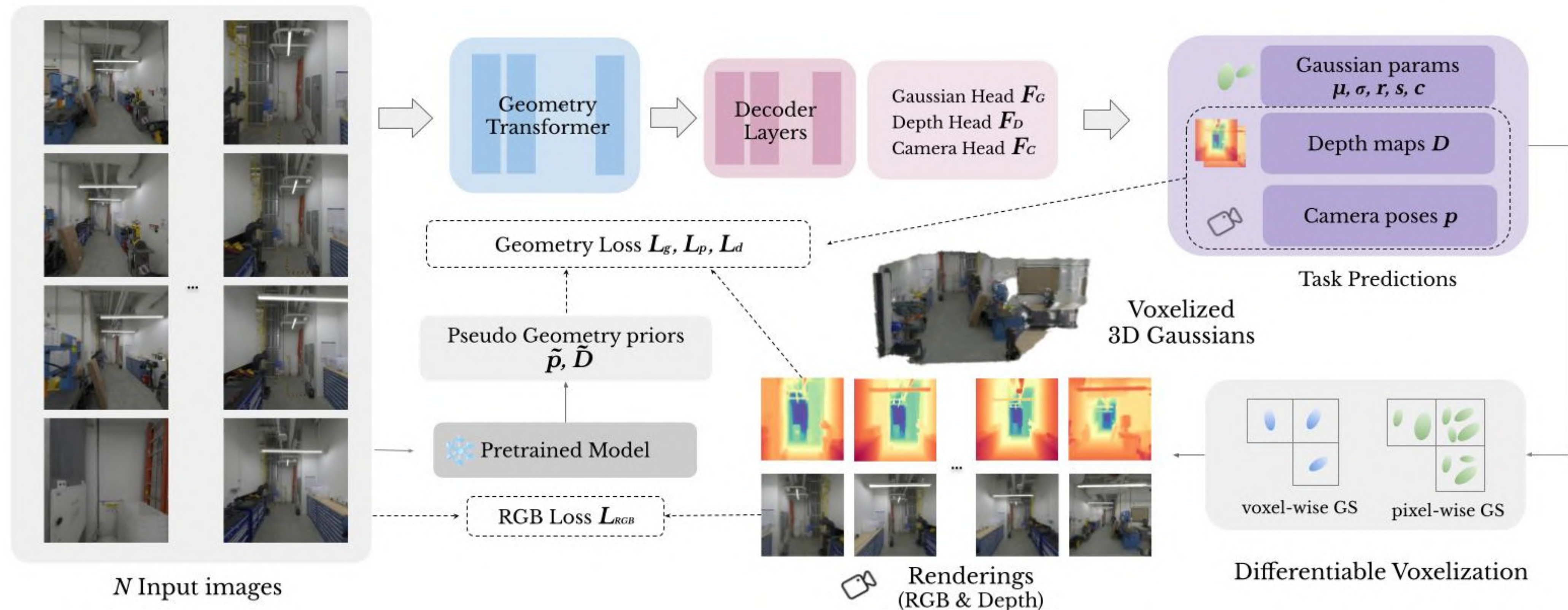


<https://instantsplat.github.io/>

<https://splatt3r.active.vision/>

AnySplat & Depth Anything 3

VGGT Successor with a Dedicated Splatting Module



Key Takeaways

- Gaussian Splatting
 - Volumetric model
 - Efficient rendering based on rasterisation
- Capable of real-time rendering
- Almost reaches visual fidelity of NeRFs
- Next time: generating 3D content with generative models

